

Law of a Random Variable

Definition

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let

$$X : \Omega \rightarrow E$$

be a random variable taking values in a measurable space $(E, \mathcal{E})$. The law, or distribution, of X is the probability measure on E defined by

$$\mathcal{L}(X)(A) = \mathbb{P}(X \in A) = \mathbb{P}(X^{-1}(A)), \quad A \in \mathcal{E}.$$

Equivalently,

$$\mathcal{L}(X) = X_{\#}\mathbb{P}.$$

Time-Dependent Random Variables

If $(X_t)_{t \geq 0}$ is a time-dependent random variable or stochastic process, then the law of X_t is

$$\mathcal{L}(X_t) = (X_t)_{\#}\mathbb{P}.$$

It describes the probability distribution of the random state at time t .